

t-SNE Project Warmup Exercise

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October 21, 2024

1 Setup

This document is dedicated to the reconstruction of [1] with low-dimension kernel g replaced by a more general kernel. This would serve as a great way to understand the material and help with the completion of the original project. For the sake of simplicity, we will use the exact same notation as in [1] and skip all the definitions. Recall in [1], g is defined as

$$g(y, y') = \frac{1}{1 + \|y - y'\|^2} \quad (1.1)$$

With all these being said, we will first start by giving our general low dimension kernel g some general assumptions, and modify it as we move on. We first require that $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' > \infty \quad (1.2)$$

We will kick off this warmup by listing all the important lemmas and results in order, and prove our own versions when the need arises. The final goal is to prove a same set of results as in [1] with the y kernel replaced. Recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) = & - \int \frac{\|x - x'\|^2}{2\sigma^2} \frac{\exp(-\|x - x'\|^2 / 2\sigma^2)}{\int \exp(-\|x - x'\|^2 / 2\sigma^2) d\mu(x')} d\mu(x') \\ & - \log \left(\int \exp(-\|x - x'\|^2 / 2\sigma^2) \right) + \log \rho. \end{aligned} \quad (1.3)$$

Proposition 1. *If μ has a C^1 density f such that both f and $\|\nabla f\|$ are bounded, then the equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. This proposition is about the kernel of x . The proof stays unchanged from [1].

□

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References

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- [3] Pavlin G. Poličar and Blaž Zupan. Visualizing high-dimensional temporal data using direction-aware t-sne, 2024, 2403.19040. URL <https://arxiv.org/abs/2403.19040>.
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